

Семинар 20 фев

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13:28

Изменение энтропии в необратимых процессах. Термодинамические потенциалы.

$$dS(T, V) = T \underbrace{\left(\frac{\partial S}{\partial T}\right)_V}_{C_V} \frac{dT}{T} + \left(\frac{\partial S}{\partial V}\right)_T dV$$

$$dU = T dS - P dV$$

$$T = \left(\frac{\partial U}{\partial S}\right)_V \quad \frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}$$

$$-P = \left(\frac{\partial U}{\partial V}\right)_S \quad \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

$$d(U - TS) = dF = -S dT - P dV$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

$$dS(T, V) = T \left(\frac{\partial S}{\partial T}\right)_V \frac{dT}{T} + \left(\frac{\partial P}{\partial T}\right)_V dV = \gamma C_V \frac{dT}{T} + \gamma \frac{R}{V} dV$$

$$S(T) = S(T_0) + C_V \ln \frac{T}{T_0} + R \ln \frac{V}{V_0}$$

$$\Delta U = 0 \Rightarrow T = \text{const}$$

$$\Delta S = \frac{m}{M} \ln \frac{V}{V_0} = \frac{m}{M} \ln 2$$

Кулёв

3.2.

$$Q_1 = 29700 \text{ Дж}$$

$$Q_2 = c_{m, \text{H}_2\text{O}} T_0 = 29700 \text{ Дж}$$

$$\Delta S_k = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = \frac{Q_1}{T_0} = \Delta S = 16,1 \frac{\text{Дж}}{\text{К}}$$

$$\Delta S_k = C_V \ln \frac{T_1}{T_0}$$

Единицы

УЧ. 75

$$dS = C_V \frac{dT}{T} + R \frac{dV}{V} \quad \ominus$$

$$dV = R \frac{dT}{P} - \frac{RT}{P^2} dP$$

$$\ominus C_V \frac{dT}{T} + \frac{R}{V} \left(R \frac{dT}{P} - \frac{RT}{P^2} dP \right) = C_V \frac{dT}{T} + \frac{R}{T} dT - R \frac{dP}{P} =$$

$$= \underbrace{(C_V + R)}_{C_P} \frac{dT}{T} - R \frac{dP}{P}$$

$$\Delta S = C_P \ln \frac{T}{T_0} - R \ln \frac{P}{P_0}$$

$$1) \Delta S_1 = C_P \ln \frac{T_2}{T} = R \ln 2$$

$$\Delta U = -2 P \Delta V = C_V (T_2 - T) = -2P(V_2 - V)$$

$$2) PV_2 = RT_2$$

$$T_2 = \frac{C_V T + 2PV}{C_V + R} = \frac{7}{5} T$$

$$\Delta S_1 = C_P \ln \frac{7}{5} - R \ln 2 = R \left(\frac{3}{2} \ln \frac{7}{5} - \ln 2 \right)$$

$$2) \left[\overline{2P, V_2} \right] \longrightarrow \left[\overline{P, V_1} \right]$$

$$\Delta S_2 = C_V \ln \frac{T_1}{T_2} + R \ln \frac{V_1}{V_2} =$$

$$= C_V \ln \frac{T_1}{T_2} + R \ln 2 \frac{T_1}{T_2} = C_P \ln \frac{T_1}{T_2} + R \ln 2$$

$$C_V (T_1 - T_2) = P(V_2 - V_1) = \frac{RT_2}{2} - RT_1$$

$$T_2 = \frac{5}{4} T_1$$

$$\Delta S_2 = \frac{5}{2} R \ln \frac{5}{4} + R \ln 2$$